Pre-flight card: *get the illumination right first*

Everything the objective is capable of is decided before you touch the camera. An objective run without Köhler illumination does not reach its rated resolution, and no amount of processing afterwards recovers what the condenser never delivered. This takes about ninety seconds. Do it once per objective, every session.

Köhler illumination

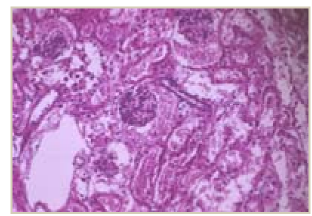
SIDE 1 · EVERY OBJECTIVE, EVERY SESSION

The wording of these six steps is Dr. Burghardt's, from the workshop lecture. The photographs are the ones he uses: what you should actually see down the eyepieces at each stage.

1 Focus the specimen in brightfield

Select your objective and focus critically. While you are here, set the eyepiece inter-pupillary distance and the diopter correction for your own eyes — do it now, not later, because everything you judge by eye from here depends on it.

> LOOK FOR — a dark shadow in the upper right of the field. That shadow is the misalignment you are about to remove. Note where it is.



Step 1 — before alignment

2 Close the field diaphragm partially

Close it down until a polygon of light sits well inside the field of view. It will be soft-edged and probably off to one side. Both of those are what the next two steps fix.

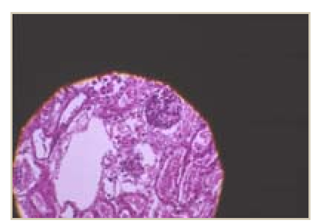
> LOOK FOR — the diaphragm edge visible but blurred. If you cannot see any edge at all, you have not closed it far enough.

Step 2 — closed down

3 Focus the diaphragm edge with condenser height

Raise or lower the condenser — not the focus knob — until the edge of the diaphragm is crisp and in the same plane as your specimen. If the polygon wanders out of the field while you do this, skip ahead to step 4, centre it, then come back here.

> LOOK FOR — specimen and diaphragm edge sharp at the same time. That is the whole condition: the field diaphragm is now conjugate with your specimen.



Step 3 — edge focused, off centre

4 Centre it with the two condenser screws

Use the two centring screws on the condenser mount. Work one at a time; they are not independent axes on most stands, so expect to go back and forth once or twice.

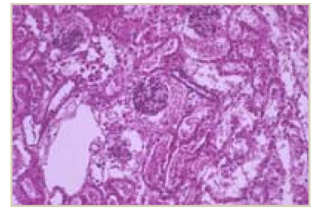
> LOOK FOR — the polygon centred in the field with a crisp edge all the way round. Uneven sharpness around the rim usually means the condenser is tilted, not off centre.

Step 4 — centred, crisp edge

5 Open the field diaphragm to the edge of the field

Open it just until the polygon disappears past the boundary of what you can see, and no further. Opening beyond that adds only stray light, which lifts your background and costs you contrast for nothing.

> LOOK FOR — the shadow you noted in step 1 is gone. If it is still there, the condenser height or centring is not yet right; go back to step 3.



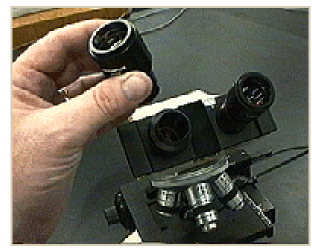
Step 5 — evenly lit, shadow gone

6 Set contrast with the condenser diaphragm — while watching the back focal plane

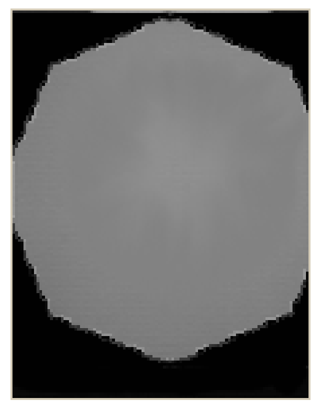
This one is different from the others, and it is the step people skip. The condenser diaphragm — also called the aperture diaphragm, which is a genuine source of confusion — controls the *angle* of the illuminating cone, not the size of the lit area. Remove one eyepiece and look down the tube, or swing in the Bertrand lens if the stand has one. You are now looking at the objective's back focal plane, and you can see the diaphragm directly.

Close it to roughly three quarters of the pupil diameter. Closing further buys apparent contrast by throwing away the outer part of the aperture, which is exactly where your finest detail is carried. That trade is almost never worth it, and it is not recoverable afterwards.

> LOOK FOR — a bright disc filling about three quarters of the pupil. If you want to understand what else is living in that disc, open Companion 01 and compare.



Step 6 — eyepiece out



what you see down the tube

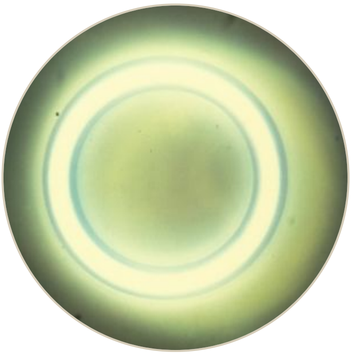
Repeat the entire process for every objective you use. Changing objective changes the aperture the condenser has to fill, so a setup that was correct at 20× is wrong at 63×. Done properly, this is what lets the lens reach its rated resolution — which is the only reason any of the rest of it matters.

INSTRUMENT	OBJECTIVE	NA	IMMERSION	DATE CHECKED

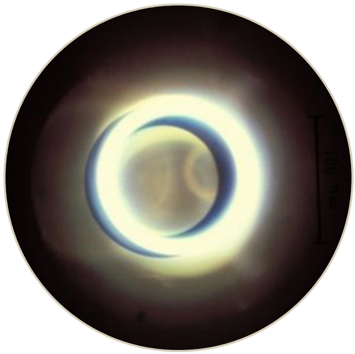
Phase contrast alignment

SIDE 2 · AFTER KÖHLER, NEVER BEFORE

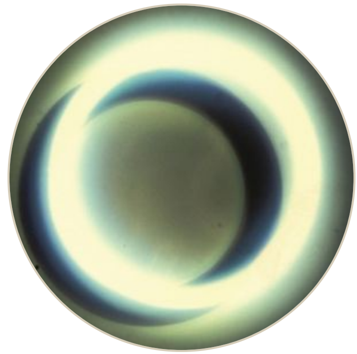
Phase contrast will produce a picture whether or not it is aligned, which is precisely the danger — a misaligned setup looks like a slightly disappointing image rather than like a fault. These four photographs were taken down the tube of a real microscope and are the states you are choosing between.



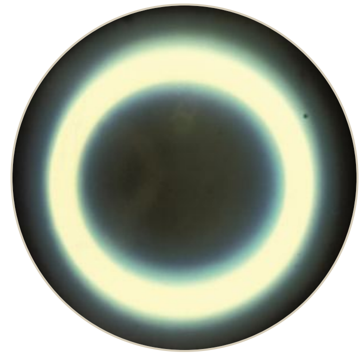
Annulus wider than the phase ring. Wrong condenser turret position for this objective.



Undersized and off centre. Very little of the ring lands on the plate.



Right size, displaced. Correct with the condenser centring screws.



Concentric and superimposed. This is the target — align to this and stop.

1 Set Köhler first, in brightfield

Run side 1 to completion with the condenser turret on its clear brightfield position. Every alignment below assumes the condenser is already at the right height and centred.

2 Match the annulus to the objective

Rotate the condenser turret to the annulus that matches the phase ring engraved on the objective — Ph1, Ph2, Ph3. The objective tells you which; the numbers are not interchangeable between manufacturers or even between objective series.

3 Look at the back focal plane

Eyepiece out, or Bertrand lens in. You should see a bright ring (the condenser annulus) and a darker ring (the phase plate in the objective). Compare what you see with the four photographs above before you touch anything.

4 Superimpose them, then put the eyepiece back

Use the phase centring screws — separate from the Köhler centring screws on most condensers — until the bright ring sits concentrically on the dark ring. Open the condenser diaphragm fully: in phase contrast the annulus defines the illumination, so closing the diaphragm only clips it.

> LOOK FOR — the bright ring covered by the dark ring all the way round. If one side stays bright, you are off centre, not the wrong size.

The image has a bright halo around every object

Some halo is intrinsic to phase contrast and cannot be removed — it is the price of the technique. Excessive halo usually means the annulus is oversized for the plate, or the specimen is far thicker than phase contrast assumes.

Contrast is poor and the background is bright

Most often misalignment. Check the back focal plane before changing anything else.

Contrast looks inverted

Wrong turret position, or a negative-phase objective. Confirm the engraving on the lens.

Everything went dim when I closed the condenser diaphragm

You are clipping the annulus. In phase contrast the diaphragm stays open.

Companion 01, the back focal plane explorer, is the interactive counterpart to step 6 and to the whole of side 2. It shows the same plane you are looking at down the tube, with the diffraction orders made visible, and lets you sweep the aperture continuously in a way the instrument cannot. Read it before the session; use this card during it.

Draft v0.1 — not yet reviewed. The six Köhler steps follow Dr. Burghardt's lecture wording closely. The "look for" notes, the phase contrast card, and the captions on the four photographs are reconstructions and need his review before this is distributed.